

Leibnitz kuralı

$$\lim_{x \rightarrow 2} \frac{1}{x-2} \int_{2x}^{x^2} \frac{1}{1+t^3} dt$$

$$\frac{\int_{2x}^{x^2} \frac{1}{1+t^3} dt}{x-2}$$

$$\frac{\frac{1}{1+t^3} \cdot 2x - \frac{1}{1+t^3} \cdot 2}{1} = \frac{4}{65} - \frac{1}{65} = \frac{3}{65}$$

limitinin değeri aşağıdakilerden

hangisidir?

A) 0

B) 1

C) ∞

D) $\frac{2}{65}$

E) $\frac{4}{65}$

1

$$h(x) = 8 + \int_1^{x^2} \frac{dt}{\cos^2(t-1)}$$

fonksiyonunun $x=1$ noktasında

(1, 8)

$$8 \int_1^{x^2} \sec^2(t-1) dt = 8 \left(\tan(t-1) \right) \Big|_1^{x^2}$$

lineer ifadesi aşağıdakilerden hangisidir?

$$8 + \tan(x^2-1) - \tan 0$$

A) $L(x) = 2x + 6$

B) $L(x) = 2x + 10$

C) $L(x) = 2x + 8$

D) $L(x) = x + 6$

E) $L(x) = x + 8$

$$\frac{1}{\cos^2(x^2-1)} \cdot 2x - \frac{1}{\cos^2 0} \cdot 0$$

2

3

$$\int_1^{\ln x} f(t) dt = e^x \sin(\ln x) \text{ ise } f(0) \text{ değeri nedir? } (x > 0)$$

$$f(\ln x) \cdot \frac{1}{x} = f(0) \cdot 0 = e^x \sin(\ln x) + e^x \cos(\ln x) \cdot \frac{1}{x}$$

A) 0

B) e

C) e^2

D) 1

E) $\sin 1$

$$f(0) = e$$

4

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^3} \int_0^x \frac{e^{t^4} - e^{-t^4}}{t^2} dt \right) = ?$$

$$\frac{8}{12} = \frac{2}{3}$$

$$\frac{\frac{e^{x^4} - e^{-x^4}}{x^2}}{3x^2}$$

$$\frac{e^{x^4} - e^{-x^4}}{3x^4} = \frac{e^{x^4} \cdot 4x^3 + e^{-x^4} \cdot 4x^3}{12x^4}$$

A) $\frac{1}{3}$

B) 1

C) $\frac{2}{3}$

D) $-\frac{1}{3}$

E) $-\frac{2}{3}$

5 $\int \tan^2 x + 1 - (\cot^2 x + 1) = \int \sec^2 x - \csc^2 x = \tan x + \cot x + c$
 $\int (\tan^2 x - \cot^2 x) dx$ integrali aşağıdakilerden hangisidir?

- A) $\sec x + \operatorname{cosec} x + c$ B) $-\tan x + \cot x - 2x + c$ C) $\tan x - \cot x + 2x + c$
D) $\tan x + \cot x + c$ E) $\cot x - \operatorname{cosec} x + c$

6 $\int (1 + \ln x + \ln(x^2 - 1) - \ln(x^3 - x)) dx$ integralinin değeri hangisidir?
 $\ln\left(\frac{x^3 - x}{x^3 - x}\right) = \ln 1 = 0 \quad \int 1 dx = x + c$

- A) c B) $\frac{1}{x} + \frac{2x}{x^2 - 1} - \frac{(3x^2 - 1)}{x^3 - x} + c$ C) $x \ln x - x \ln(x^2 - 1) + c$ D) $x + c$

7 $\int \left(\frac{dx}{4 + 9x^2} \right)$ integralinin değeri aşağıdakilerden hangisidir?

$\int \frac{dx}{4(1 + \frac{9x^2}{4})} \quad \frac{1}{4} \int \frac{dx}{1 + (\frac{3x}{2})^2} = \frac{1}{4} \cdot \frac{2}{3} \arctan \frac{3x}{2} = \frac{1}{6} \arctan \frac{3x}{2} + c$

- A) $\frac{1}{3} \ln \left| \frac{9x}{4} \right| + c$ C) $\frac{1}{6} \ln \left| \frac{3x}{2} \right| + c$ E) $\frac{1}{6} \arctan \frac{3x}{2} + c$
B) $\ln |x| + c$ D) $\arcsin \frac{3x}{2} + c$
C) $\frac{1}{6} \ln \left| \frac{3x}{2} \right| + c$
D) $\arcsin \frac{3x}{2} + c$

8 $\int e^{-2 \ln(\sqrt{x})} dx = ? \quad e^{\ln(\sqrt{x})^2} \quad \int \frac{1}{x} dx = \ln x + c$

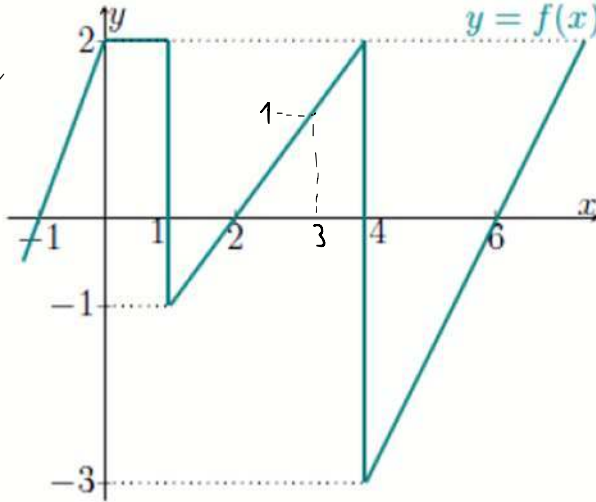
- A) c B) $\ln x + c$ C) $x + c$ D) $\frac{1}{\sqrt{x}} + c$ E) $\frac{1}{\sqrt[3]{x^2}} + c$

9 $\int_0^3 g(x) dx = 7, \int_0^6 g(2x) dx = 5 \Rightarrow \int_0^2 x g(3x^2) dx = ?$
 $\int_0^2 x g(3x^2) dx = ?$
 $2x = u \quad 2dx = du \quad \int_0^2 x g(3x^2) dx = \int_0^2 \frac{u}{6} g(u) du = \frac{1}{6} \int_0^2 u g(u) du$
 $\int_0^2 u g(u) du = 10 \quad \frac{1}{6} \cdot 10 = \frac{5}{3}$
a) 2 b) $\frac{1}{2}$ c) 10 d) 3 e) $\frac{5}{6}$ $\frac{3}{6} = \frac{1}{2}$

10 $\int \frac{\arctan(\ln x)}{x(1 + \ln^2 x)} dx = ?$
 $\arctan(\ln x) = u \quad \frac{1}{1 + \ln^2 x} \cdot \frac{1}{x} dx = du \quad \int \frac{u}{1 + u^2} du$
 $10 - 7 = 3 \quad \frac{u^2}{2} + c$
A) $\arctan x + c$ B) $\frac{1}{2} \arctan(\ln x) + c$ C) $x \arctan(\ln x) + c$
D) $\frac{1}{2} (\arctan(\ln x))^2 + c$ E) $\arctan(\ln x) + \arctan x + c$
 $\frac{\arctan^2(\ln x)}{2} + c$

11) Aşağıda grafiği verilen $y = f(x)$ fonksiyonu için

$$\int_{-1}^{1/2} f(x) dx + \int_3^6 f(x) dx + \int_{1/2}^3 f(x) dx = ?$$



$$1 + 1 - 3 + \frac{3}{2} + 1$$

$$1 \frac{1}{2}$$

$$2 - \frac{1}{2} = \frac{3}{2}$$

$$2 \frac{3}{2}$$

- A) $\frac{3}{2}$ B) $\frac{5}{2}$ C) $\frac{9}{2}$ D) $\frac{11}{2}$ E) $\frac{17}{2}$

12) f fonksiyonu $[0, 4]$ aralığında tanımlı sürekli bir fonksiyon ve ters türevi de F olmak üzere

$$\int_0^4 f(x) dx = 10! \text{ ve } F(0) = 8! \text{ ise } F(4) = ? \quad F(4) - F(0) = 10! \quad F(4) - 8! = 10!$$

- A) $10! + 8!$ B) $10! - 8!$ C) 0 D) $2!$ E) $4!$

$$F(4) = 8! (1 + 90)$$

13) $\begin{cases} F(x) = (f \circ g)(x) \\ g(2) = \frac{\pi}{4}; g(-1) = \frac{\pi}{3} \\ f(x) = \tan x \end{cases}$ bilgileri veriliyor. F fonksiyonu h fonksiyonunun bir ters türevi

$$\text{olduğuna göre } \int_{-1}^2 h(x) dx = ?$$

$$F(2) - F(-1)$$

$$1 - \sqrt{3}$$

- A) $\sqrt{2} - \sqrt{3}$ B) $\sqrt{3}$ C) $1 - \sqrt{3}$ D) 1 E) $\sqrt{3} - \sqrt{2}$

14) $f(x) = \begin{cases} \sin x \cos x, & 0 \leq x \leq \frac{\pi}{2} \\ \cos^2 x, & \frac{\pi}{2} < x \leq \pi \end{cases}$ fonksiyonu için $\int_0^\pi f(x) dx$ belirli integralinin

değeri kaçtır?

$$\int_0^{\frac{\pi}{2}} \sin x \cos x dx + \int_{\frac{\pi}{2}}^\pi \cos^2 x dx$$

$$\cos 2x = 2 \cos^2 x - 1$$

$$\frac{\cos 2x + 1}{2}$$

$$\frac{1}{2} + \frac{\pi}{4}$$

$$A) 1$$

$$C) \frac{\pi}{4}$$

$$E) \frac{\pi+1}{2}$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{2} dx + \int_{\frac{\pi}{2}}^\pi \frac{\cos 2x}{2} + \frac{1}{2} dx$$

$$B) \frac{\pi}{2}$$

$$D) \frac{\pi}{4} + \frac{1}{2}$$

$$\left[-\frac{\cos 2x}{4} + \frac{\sin 2x}{4} + \frac{x}{2} \right]_{\frac{\pi}{2}}^\pi = \left[-\frac{\cos 2\pi}{4} + \frac{\sin 2\pi}{4} + \frac{\pi}{2} \right] - \left[-\frac{\cos \pi}{4} + \frac{\sin \pi}{4} + \frac{\pi}{4} \right] = \left[-\frac{1}{4} + 0 + \frac{\pi}{2} \right] - \left[\frac{1}{4} + 0 + \frac{\pi}{4} \right] = \frac{\pi}{2} - \frac{1}{4} - \frac{\pi}{4} - \frac{1}{4} = \frac{\pi}{4}$$

15) $\int_{-\pi/4}^{\pi/4} [\tan^3(x^3) + \tan^2 x] dx$ belirli integralinin değeri kaçtır?

A) 0

B) $\sqrt{2} - \frac{\pi}{2}$

C) $2 - \frac{\pi}{2}$

D) π

E) $1 + \frac{\pi}{2}$

$$\int_{-\pi/4}^{\pi/4} \tan^3(x^3) dx + \int_{-\pi/4}^{\pi/4} \tan^2 x dx$$

$\tan x^3 = u$

$$\frac{\sec^2 x^3}{3x^2} dx = du$$

$$(\sec^2 x^3 - 1) \cdot u dx$$

$$\sec^2 x - 1$$

$$\tan x - x \Big|_{-\pi/4}^{\pi/4}$$

$$1 - \frac{\pi}{4} + 1 - \frac{\pi}{4}$$

$$2 - \frac{\pi}{2}$$

16) $\int_{-2}^2 \left(\frac{x^3 \cos^3 x}{9} - \frac{x}{7} - \frac{x^5 \sin^2 x}{7} + \frac{x^2 \sin x}{4} + 3 \right) dx = ?$

A) 12

B) 0

C) 2

D) 4

E) 8

$$3 \times \int_{-2}^2 2 dx = 6 + 6 = 12$$

17) Aşağıda verilen belirsiz integraller ile çözümleri için yapılabilecek dönüşümler hangi seçenekte doğru olarak eşleştirilmiştir?

1. $\int \frac{dx}{2 - \sin x}$

a. $x = \frac{\sqrt{7}}{2} \tan t - \frac{1}{2}$

2. $\int \frac{dx}{\sqrt{x^2 + x + 2}}$

b. $x = \frac{\sqrt{2}}{3} \sin t$

$$\sqrt{2} \cos t = \sqrt{2} \cdot 9x$$

3. $\int \sqrt{2 - 9x^2} dx$

c. $x = 3 \sec t + 1$

$$\frac{\sqrt{2}}{\sqrt{2-9x^2}} \cdot 3x$$

$$\sqrt{2} \sin t = 3x$$

$$\sqrt{2} \cos t dt = 3 dx$$

4. $\int \frac{dx}{\sqrt{x^2 - 2x - 8}}$

d. $\tan \frac{x}{2} = t$

$$\frac{2}{3} \cos^2 t \cdot \frac{\sqrt{2}}{3} \cos t dt = dx$$

A) $\frac{1-a}{2-b}$
 $\frac{3-c}{4-d}$

B) $\frac{1-d}{2-a}$
 $\frac{3-b}{4-c}$

C) $\frac{1-c}{2-b}$
 $\frac{3-d}{4-a}$

D) $\frac{1-d}{2-a}$
 $\frac{3-c}{4-b}$

E) $\frac{1-d}{2-c}$
 $\frac{3-a}{4-b}$

18) Aşağıda verilen belirsiz integraller ile çözümleri için yapılabilecek dönüşümler hangi seçenekte doğru olarak eşleştirilmiştir?

1. $\int \frac{dx}{2-\cos x}$

a. $x = \frac{\sqrt{3}}{2} \tan t - \frac{1}{2}$

2. $\int \frac{dx}{\sqrt{x^2+x+1}}$

b. $\tan \frac{x}{2} = t$

3. $\int \sqrt{4-3x^2} dx$

$\begin{array}{c} 2 \\ \sqrt{4-3x^2} \end{array} \quad \begin{array}{c} \sqrt{3}x \\ 2 \sin t = \sqrt{3}x \end{array}$

c. $x = 2 \sec t + 1$

4. $\int \frac{dx}{\sqrt{x^2-2x-3}}$

$\sqrt{(x-1)^2-4}$

$\begin{array}{c} x-1 \\ \sqrt{(x-1)^2-4} \end{array} \quad \begin{array}{c} 2 \\ \sqrt{4-3x^2} \end{array}$

d. $x = \frac{2\sqrt{3}}{3} \sin t$

$\cos t = \frac{2}{x-1} \quad 2 \sec t = x-1$

A) $\begin{array}{c} 1-a \\ 2-b \\ 3-c \\ 4-d \end{array}$

B) $\begin{array}{c} 1-b \\ 2-a \\ 3-d \\ 4-c \end{array}$

C) $\begin{array}{c} 1-c \\ 2-b \\ 3-d \\ 4-a \end{array}$

D) $\begin{array}{c} 1-b \\ 2-d \\ 3-c \\ 4-a \end{array}$

E) $\begin{array}{c} 1-b \\ 2-a \\ 3-c \\ 4-d \end{array}$

Buradan sonrakiler kontrol edilmedi!

19) $\int_{-1}^1 (x^3 \cos x + \sin^7 x - x^2 \sin x + x^4) dx = ?$ A) $\frac{\cos 1}{3} + \frac{4}{5}$ B) $\frac{2}{5}$ C) $\frac{\cos 1}{3} - \frac{4}{5}$ D) $\frac{3}{5}$ E) 0

$\frac{x^5}{5} \Big|_{-1}^1 = \frac{1}{5} - \frac{-1}{5} = \frac{2}{5}$

$\int \left(\frac{2x}{x^2+3} - e^{5x} + \sin 4x - \frac{1}{x^2+9} \right) dx = ?$

$\frac{e^{5x}}{5} - \frac{\cos 4x}{4} - \frac{1}{9} \cdot \frac{1}{1+\frac{1}{3}} \cdot \frac{-1}{3} \arctan \frac{x}{3}$

A) $\ln(x^2+3) - \frac{e^{5x}}{5} - \frac{\cos 4x}{4} - \frac{1}{3} \arctan \frac{x}{3} + c$

B) $\frac{1}{\sqrt{3}} \arctan \frac{x}{\sqrt{3}} - \frac{e^{5x}}{5} + 4 \cos 4x - \ln(x^2+9) + c$

C) $\ln(x^2+3) - \frac{e^{5x}}{5} - 4 \cos 4x - \arctan \frac{x}{3} + c$

D) $\frac{1}{\sqrt{3}} \arctan \frac{x}{\sqrt{3}} - 5e^{5x} + 4 \cos 4x - \ln(x^2+9) + c$

E) $\ln(x^2+3) - 5e^{5x} - \frac{\cos 4x}{4} - \arctan \frac{x}{3} + c$

20) $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin(t^2) dt}{2x^6} = ?$ A) ∞ B) 0 C) $\frac{1}{4}$ D) $\frac{1}{2}$ E) $\frac{1}{6}$

$\frac{\sin(x^4) \cdot 2x}{12x^5} = \frac{\sin(x^4)}{6x^4}$

$\frac{\cos(x^4) \cdot 4x^3}{24x^3} = \frac{\cos(x^4)}{6}$

$\frac{1}{6} \cos x^4$

(21) $\int \frac{dt}{t^2(t^4-16)}$ integralinin basit kesirlere ayrılmış hali

$t^2, (t^2-4), (t^2+4)$

$t^2(t-2)(t+2)(t^2+4)$

aşağıdakilerden hangisi olabilir?

A) $\int \left(\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t+2} + \frac{D}{(t-2)^2} + \frac{Et+F}{t^2+4} \right) dt$

B) $\int \left(\frac{A}{t} + \frac{B}{t-2} + \frac{C}{t+2} + \frac{Dt+E}{t^2+4} \right) dt$

C) $\int \left(\frac{A}{t} + \frac{B}{t-2} + \frac{Ct+D}{(t-2)^2} + \frac{E}{t+2} + \frac{Ft+G}{t^2+4} \right) dt$

D) $\int \left(\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t-2} + \frac{D}{t+2} + \frac{Et+F}{t^2+4} \right) dt$

E) $\int \left(\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t+2} + \frac{Dt+E}{t^2+4} \right) dt$

$\cos(\pi x^2)$

$\cos \frac{\pi}{6}$

$\cos 30$

(22) $F(x) = \int_0^x \cos(\pi t^2) dt$ eğrisinin $x = \frac{1}{\sqrt{6}}$

noktasındaki teğetin eğimi aşağıdakilerden hangisidir?

A) $\frac{\sqrt{3}}{2}$ B) 1 C) $\frac{\pi}{\sqrt{3}} - 1$ D) $1 - \frac{1}{\sqrt{3}}$ E) $\sqrt{3} - 1$

(23) $\int_{\pi/4}^{\pi/2} (\sin x)^3 dx = ?$

$\sin^2 x$

$\cos x = u$

$-\sin x dx = du$

$-\int_{\pi/4}^{\pi/2} u^2 du$

$-\left(\cos x - \frac{\cos^3 x}{3} \right) \Big|_{\pi/4}^{\pi/2}$

$\frac{2\sqrt{2}}{24}$

A) 1 B) $\frac{\sqrt{3}}{8}$ C) $\frac{2}{3} - \frac{5\sqrt{2}}{12}$ D) $\frac{2}{3} - \frac{\sqrt{3}}{8}$ E) $\frac{5\sqrt{2}}{12}$

$-\left(u - \frac{u^3}{3} \right) \Big|_{\pi/4}^{\pi/2}$

$\left(\cos 90 - \frac{\cos^3 90}{3} \right) - \cos 45 + \frac{\cos^3 45}{3}$

$-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{12}$

$-\frac{6\sqrt{2}}{12} + \frac{\sqrt{2}}{12}$

$-\frac{5\sqrt{2}}{12}$

(24) $\int_1^{e^4} \frac{\sec^2(\ln x)}{x} dx = ?$

$\ln x = u$

$\frac{1}{x} dx = du$

$\int_1^{e^4} \sec^2 u du$

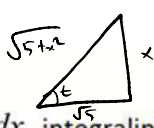
$\tan(\ln x) \Big|_1^{e^4}$

A) $e^4 - \ln \frac{\sqrt{2}}{2}$ B) 1 C) $\frac{\pi}{4} - 1$ D) $1 - \ln \sqrt{2}$ E) $\frac{\pi^3}{4^3} + \ln \sqrt{2}$

$\tan \frac{\pi}{4} - \tan 0$

$1 - 0$

25) $\int \frac{x^3}{\sqrt{5+x^2}} dx$ integralini çözebilmek için yapılması gereken trigonometrik dönüşüm ve bu dönüşüm sonrası oluşan yeni integral aşağıdakilerin hangisinde doğru verilmiştir?



$$\sqrt{5} \tan t = x$$

$$\sqrt{5} \sec^2 t dt = dx$$

$$5\sqrt{5} \cdot \tan^3 t$$

$$\sqrt{5} \cdot \sec^2 t$$

$$25 \tan t \cdot \sec^2 t$$

$$\cos t = \frac{\sqrt{5}}{\sqrt{5+t^2}} \quad \sqrt{5} \sec t$$

A) Yapılması gereken dönüşüm: $x = \sqrt{5} \tan t$, Yeni integral: $\int 5\sqrt{5} (\tan t)^3 \cdot (\sec t)^2 dt$

B) Yapılması gereken dönüşüm: $x = 5 \sin t$, Yeni integral: $\int \sqrt{5} (\sin t)^3 \cdot \cos t dt$

C) Yapılması gereken dönüşüm: $x = 5 \tan t$, Yeni integral: $\int (\tan t)^3 \cdot \sec t dt$

D) Yapılması gereken dönüşüm: $x = \sqrt{5} \sin t$, Yeni integral: $\int 5\sqrt{5} (\sin t)^3 dt$

E) Yapılması gereken dönüşüm: $x = \sqrt{5} \tan t$, Yeni integral: $\int 5\sqrt{5} (\tan t)^3 \cdot \sec t dt$

26) $\int_0^5 \operatorname{arccot} x dx = ?$ A) $25 \operatorname{arccot} 5$ B) $5 \operatorname{arccot} 5 + \ln 26$ C) $5 \operatorname{arccot} 5 + \frac{\ln 26}{2}$ D) $25 \operatorname{arccot} 5 + \frac{\ln 26}{2}$ E) $25 \operatorname{arccot} 5 + \ln 26$

$$\operatorname{arccot} x = u$$

$$dx = -du$$

$$\frac{-1}{1+x^2} dx = du$$

$$x = v$$

$$\int \frac{x}{1+x^2} dx$$

$$1+x^2 = u$$

$$2x dx = du$$

$$\frac{1}{2} \int \frac{1}{u} du$$

$$\frac{1}{2} \ln |1+x^2|$$

$$x \cdot \operatorname{arccot} x + \frac{\ln |1+x^2|}{2} \Big|_0^5$$

$$5 \operatorname{arccot} 5 + \frac{\ln 26}{2}$$